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GraphSL: An Open-Source Library for Graph Source Localization Approaches and Benchmark Datasets

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Software

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Summary

We present GraphSL, a novel library designed for investigating the graph source localization problem. Our library facilitates the exploration of various graph diffusion models for simulating information spread and enables the evaluation of cutting-edge source localization approaches on established benchmark datasets. The source code of GraphSL is made available at (https://github.com/xianggebenben/GraphSL). Bug reports and feedback can be directed to the Github issues page (https://github.com/xianggebenben/GraphSL/issues).

Statement of Need

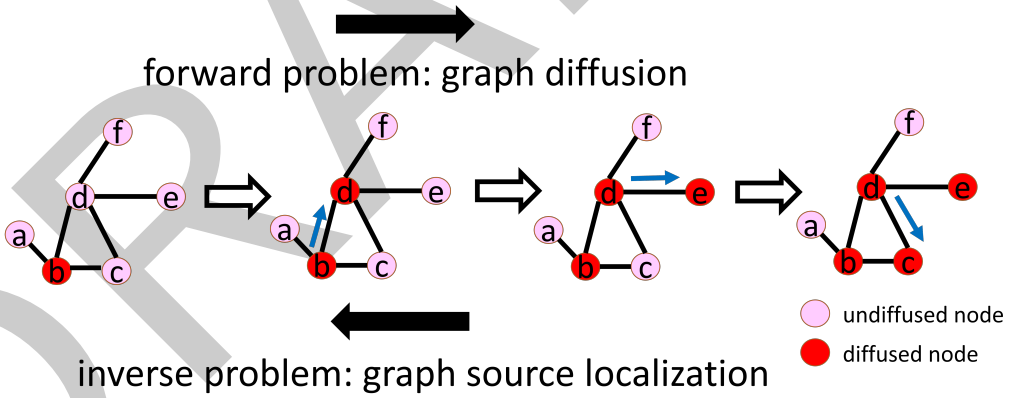

Figure 1: An example of graph source localization.

Graph diffusion is a fundamental task in graph learning, which aims to predict future graph cascade patterns given source nodes. Conversely, its inverse problem, graph source localization, though rarely explored, stands as an extremely important topic: it focuses on the detection of source nodes given their future graph cascade patterns. As illustrated in Figure 1, graph diffusion seeks to predict the cascade pattern $\{b, c, d, e\}$ from a source node b, whereas graph source localization aims to identify source nodes b from the cascade pattern $\{b, c, d, e\}$. Graph source localization spans a broad spectrum of promising research and real-world applications such as rumor detection (Evanega et al., 2020), tracking of sources for computer viruses, (Kephart & White, 1993), and failures detection in smart grids (Amin & Schewe, 2007). Hence, the graph source localization problem demands attention and extensive investigations from machine learning researchers.

Some open-source tools have been developed to support the research of the graph source localization problem due to its importance. Two recent examples are cosasi (McCabe, 2022) and

RPaSDT (Fraszczak, 2022). However, they missed comprehensive simulations of information diffusion, real-world benchmark datasets, and up-to-date state-of-the-art source localization approaches. To fill this gap, we propose a new library GraphSL: the first one to include both real-world benchmark datasets and recent source localization methods to our knowledge, which enables researchers and practitioners to easily evaluate novel techniques against appropriate baselines. These methods do not require prior knowledge (e.g. single source or multiple sources), and can handle graph source localization based on various diffusion simulation models such as Independent Cascade (IC) and Linear Threshold (LT). Our GraphSL library is standardized: for instance, tests of all source inference methods return a Metric object, which provides five performance metrics (accuracy, precision, recall, F-score, and area under ROC curve) for performance evaluation.

Methods and Benchmark Datasets

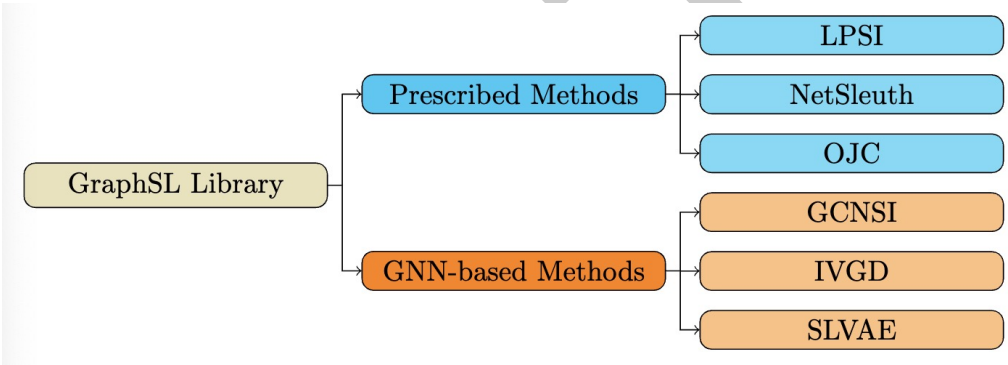

Figure 2: The hierarchical structure of our GraphSL library version 0.10.

The structure of our GraphSL library is depicted in Figure 2. Existing graph source localization methods can be categorized into two groups: Prescribed methods and Graph Neural Networks (GNN)-based methods.

Prescribed methods rely on hand-crafted rules and heuristics. For instance, LPSI propagated infection in networks and labels local peaks as source nodes (Z. Wang et al., 2017). NetSleuth employed the Minimum Description Length principle to identify the optimal set of source nodes and virus propagation ripple (Prakash et al., 2012). OJC identified a set of nodes (Jordan cover) that cover all observed infected nodes with the minimum radius (Zhu et al., 2017).

GNN-based methods learn rules from graph data in an end-to-end manner by capturing graph topology and neighboring information. For example, GCNSI utilized LPSI to enhance input and then applied Graph Convolutional Networks (GCN) for source identification (Dong et al., 2019); IVGD introduced a graph residual scenario to make existing graph diffusion models invertible, and it devises a new set of validity-aware layers to project inferred sources to feasible regions (J. Wang et al., 2022). SLVAE used forward diffusion estimation and deep generative models to approximate source distribution, leveraging prior knowledge for generalization under arbitrary diffusion patterns (Ling et al., 2022).

Table 1: The statistics of eight datasets.

Dataset	#Node	#Edge	Average Degree	Seed-Diffusion Pairs
Karate	34	78	4.588	No
Dolphins	62	159	5.129	No
Jazz	198	2,742	27.697	No

Dataset	#Node	#Edge	Average Degree	Seed-Diffusion Pairs
Network Science	1,589	2,742	3.451	No
Cora-ML	2,810	7,981	5.68	No
Power Grid	4,941	6,594	2.669	No
Memetracker	7,884	47,911	12.154	Yes
Digg	15,912	78,649	9.885	Yes

Aside from methods, we also provide eight benchmark datasets to facilitate the research of graph source localization, whose statistics are shown in Table 1. Memetracker and Digg provide seed-diffusion pairs. While others do not have such pairs, they can be generated by information diffusion simulations.

Availability and Documentation

GraphSL is available under the MIT License. The library may be cloned from the GitHub repository, or can be installed by pip: pip install GraphSL. Documentation is provided via Read the Docs, including a quickstart introducing major functionality and a detailed API reference. Extensive unit testing is employed throughout the library.

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